

Brainstorming & Idea Prioritization

Credit Card Approval Prediction System | Brainstorming & Ideation Phase

Ideas Considered

#	Idea	Description
1	Credit Card Approval Predictor (Web App)	Binary classifier (Random Forest) trained on merged application + credit records. Flask UI with
2	Credit Score Calculator	Regression model estimating a numeric credit score instead of binary approval.
3	Fraud Detection System	Anomaly detection on transaction patterns to flag fraudulent credit card usage.
4	Loan Default Risk Dashboard	Visual dashboard showing default probability trends across customer segments.
5	Customer Segmentation Tool	Clustering model grouping applicants into risk tiers for targeted marketing.

Prioritization Matrix

Idea	Impact	Effort	Data Available	Decision
1. Approval Predictor	High	Medium	Yes — 2 CSVs ready	Selected ✓
2. Credit Score Calc.	Medium	Medium	Partial	Backlog
3. Fraud Detection	High	High	No transaction data	Rejected v1
4. Default Dashboard	Medium	High	Partial	Rejected v1
5. Segmentation Tool	Low-Med	Medium	Yes	Backlog

Final Selected Idea

Idea #1 — Credit Card Approval Predictor was selected because both required datasets are available (application_record.csv and credit_record.csv), the binary classification target is straightforward to derive from STATUS codes, and the Flask web app with a clear Approved/Rejected result page creates a compelling, demonstrable product within the project timeline.